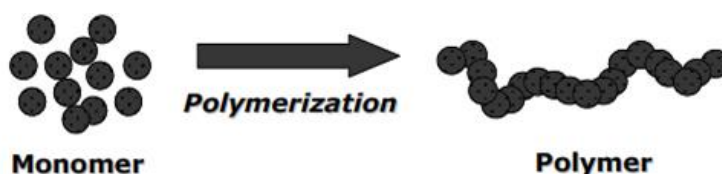


Definition: “Polymers are carbonaceous materials with large molecular masses obtained by covalent linkages of several small repeating units called monomers.

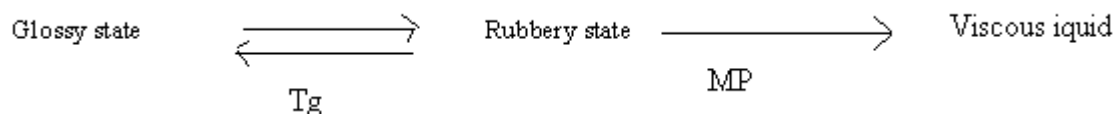
Polymerization: “Polymerization is union of two or more smaller and simpler molecules of similar or different types with or without elimination of any part of the molecule resulting in the formation of new covalent linkages.”



- GLASS TRANSITION TEMPERATURE (T_g)**

Unlike in metal article fabrication, the polymers are not melted to the liquid form to fabricate, but heated just to its softening temperature. This is referred to as glass transition temperature.

Definition of T_g : “The temperature at which a polymer abruptly transforms from glassy state (hard state) to a rubbery state (soft state) is called Glass Transition Temperature and denoted by T_g .”



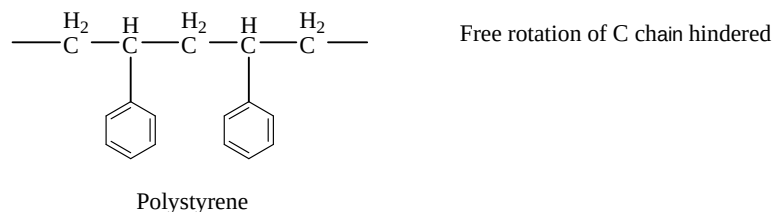
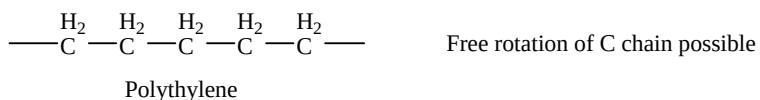
This transition from hard solid state to soft and flexible state corresponds to the beginning of chain motion or rotation about C-C single bonds in polymers. Below T_g polymer is said to be in frozen state.

Parameters affecting T_g :

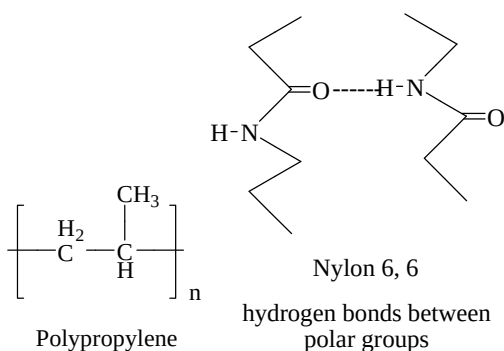
a) Chain flexibility: Free rotational motion imparts flexibility. Rigid structure in the polymer chain or with bulky side chains hinder the freedom of rotation and decrease in flexibility and increase T_g . Hence, highly flexible polymer will have low T_g value.

Ex; T_g of polyethylene is $-110\text{ }^{\circ}\text{C}$

T_g of polystyrene is $100\text{ }^{\circ}\text{C}$

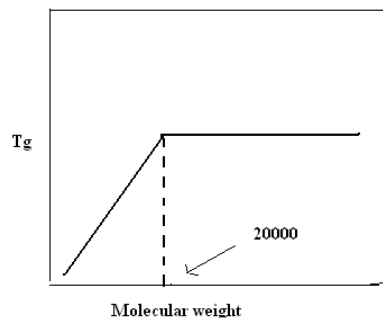


b) Intermolecular forces: Presence of large number of polar groups in molecular chain lead to strong inter molecular forces such as hydrogen bonding, Van der Waals forces etc., and they restrict molecular mobility and hence increase T_g . Hence higher the intermolecular forces higher will be the T_g .



c) Branching and Cross linking: A high degree of branching and cross linking in the polymer, brings the molecular chains closer, lowers the free volume reducing chain mobility. This will increase T_g . Hence highly branched and cross linked polymers will have higher T_g .

a) Molecular weight: High molecular weight polymers have greater density and increase in T_g . However, T_g is not significantly affected by the molecular mass if the degree of polymerization above 250 or molecular mass is above 20000. (graph)

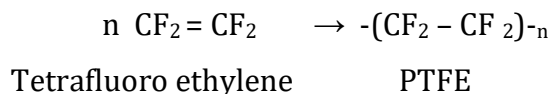


Significance of Tg:

- i) Tg can be used to evaluate the flexibility of a polymer and predict its response to mechanical stress.
- ii) Many polymers change their other physical properties like viscosity, elastic nature, etc., also at Tg. Hence, the use of a polymer at Tg with respect to other properties can be decided.
- iii) Different fabrication works like moulding, shaping, calendaring, etc., can be conveniently carried out with knowledge of Tg, as the entire machinery can be pre-set to that temperature.

• Synthesis, Properties, Applications of TEFLON (Poly tetra fluoro ethylene – PTFE)

Synthesis: It is made by addition polymerization of water emulsion of Tetra fluoro ethylene in aqueous or alkaline medium at 55-240° C in the presence of Benzoyl peroxide as catalyst at 1000 PSI pressure.



Properties: 1. High degree of crystallinity of 93-98 %, high melting point of 327° C and high density of 2.3 g cm⁻³.

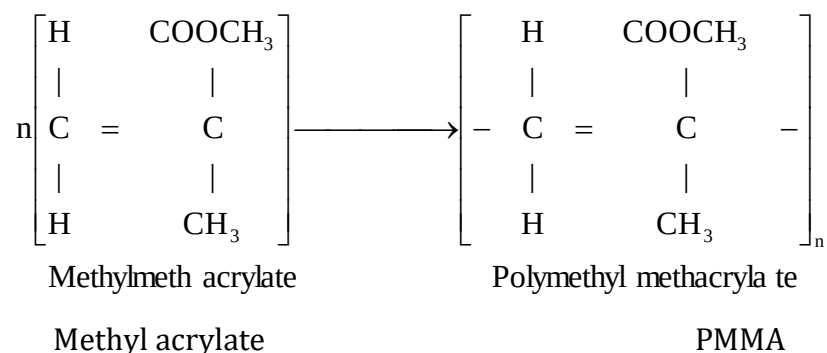
- 2. Insoluble in most solvents and chemicals
- 3. Not wetted by oil and water
- 4. High thermal stability, strength and insulating properties.

Applications: 1. For insulation in motors, transformers coil and capacitors

- 2. Making industrial filters, gaskets and belts
- 3. As non-sticky coating on bakery trays and frying pans
- 4. As anti-corrosive coating
- 5. As non-lubricated bearing.

- **Synthesis, Properties, Applications of PLEXI GLASS (Poly methyl methacrylate) PMMA**

Synthesis: it is made by bulk or emulsion polymerization of Methyl methacrylate in presence of H_2O_2 as initiator at $60-70^\circ C$ under atmospheric pressure.



Properties:

1. Highly transparent, thermoplastic, softening at $120^\circ C$.
2. Excellent optical properties
3. Not affected by sunlight but affected by many chemicals and some organic solvents
4. Has good dimensional stability.

Applications:

1. For making lenses, aircraft parts and light fixtures
2. Used in jewelry, wind screens and TV screens.
3. For making artificial eyes and denture.
4. Used in emulsion paints.

- **Conducting Polymers**

Most of the polymers are insulators, with desirable properties such as light weight, processability, durability, and low cost. By designing the molecular structures of polymers, chemists have developed new materials that exhibit electrical conductivities comparable to metals while retaining the advantages of polymers. Generally, these electrically conducting polymers are composed of conjugated polymer chains with π -electrons delocalized along the backbone. The polymers are converted to the electrically conductive, or doped, form via oxidation or reduction reactions that create delocalized charge carriers. Charge balance is accomplished by incorporating an oppositely charged counter ion into the polymer matrix.

An organic polymer with highly delocalized pi-electron system, having electrical conductance of the order of a conductor is called a conducting polymer.

An organic polymer can be converted into a conducting polymer if it has

⇒ Linear structure

⇒ Extensive conjugation in polymeric back bone (pi-back bone)

The conducting polymers are synthesized by doping, in which charged species are introduced in organic polymers having pi-back bone. The important doping reactions are:

- ⇒ Oxidative doping (p-doping)
- ⇒ Reduction doping (n-doping)
- ⇒ Protonic acid doping (p-doping)
- ⇒ Normally electron in a polymer are localized and do not take part in conducting but doping can create delocalized electrons and responsible for conduction.

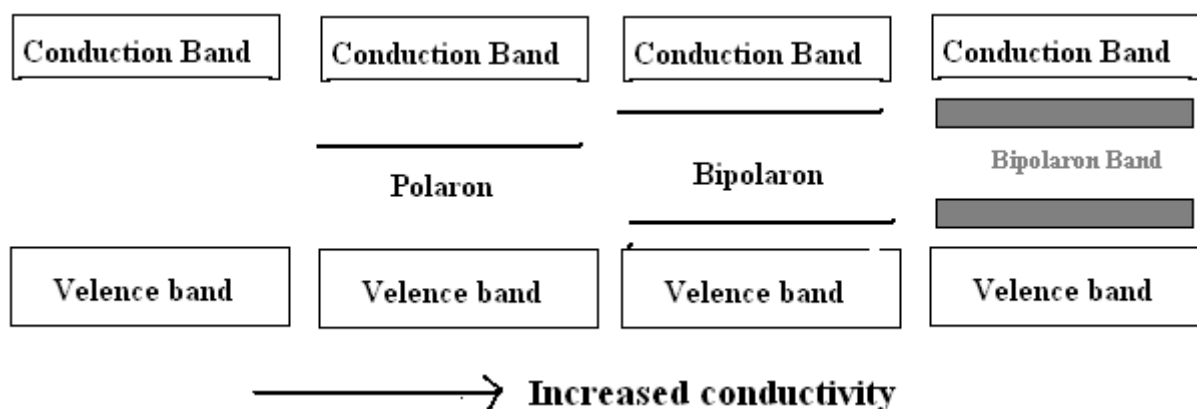
Mechanism of conduction:

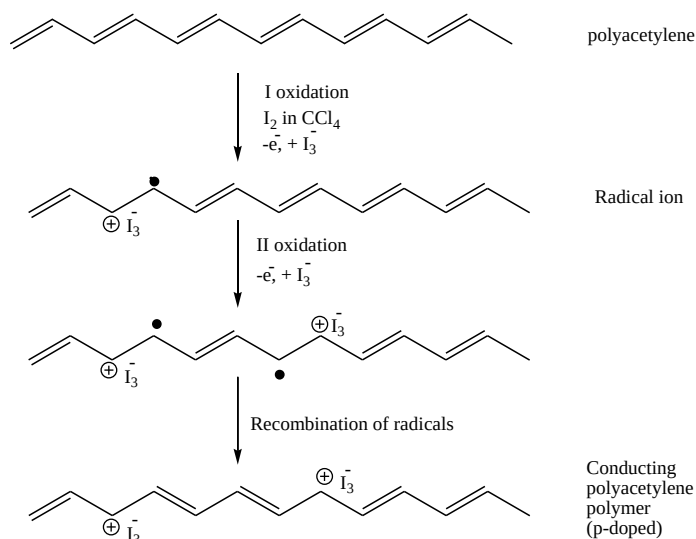
Band theory of solids has been used to explain the conduction mechanism in conducting polymers. The neutral polymer (insulator) be transferred into a conductor by doping with electron donar (reducing dopant such as sodium naphthalide) or an electron acceptor (like iodine vapor, FeCl_3 , iodine in CCl_4 , HBF_4 , perchloric acid and benoquinone)

The removal of an electron from the polymer pi-back bone using a suitable oxidizing agent leads to the formation of delocalized radical ion called polaron.

A second oxidation of a chain containing polaron, followed by radical recombination yields two charge carriers on each chain. The positive charge sites on the polymer chains are compensated by anions (I_3^-) formed by the oxidizing agent during doping.

The delocalized positive charges on the polymer chain are mobile, not the dopant anions. Thus, these delocalized positive charges are current carries for conduction. These charges must move from chain to chain as well as along the chain for bulk conduction. On doping polyacetylene using iodine in CCl_4 , for partial oxidation, the conductivity increases from $10^{-5} \text{ S. cm}^{-1}$ to $10^3\text{-}10^5 \text{ S.cm}^{-1}$.





Liquid crystals

Q- What are liquid crystals? Explain molecular ordering positional and orientational order in solid, liquid and liquid crystals. Mention general applications of liquid crystals. Definition: Liquid crystals (LCS) may be defined as *the distinct state of matter, in which the degrees of molecular ordering (positional and orientation order) is lies between ordered anisotropic crystalline state and completely disordered isotropic liquid state.*

The terms *mesophase* and *mesogen* also refer to the liquid crystal state. Liquid crystals exhibit optical anisotropy, i.e., they possess different optical properties when light is incident in different directions. Liquids, however, exhibit optical isotropy i.e., they exhibit the same optical property irrespective of the direction of the incident light. All LCS are organic molecules with *rod* or *disk* shape.

Positional and orientational order in solids, liquids and liquid crystals :

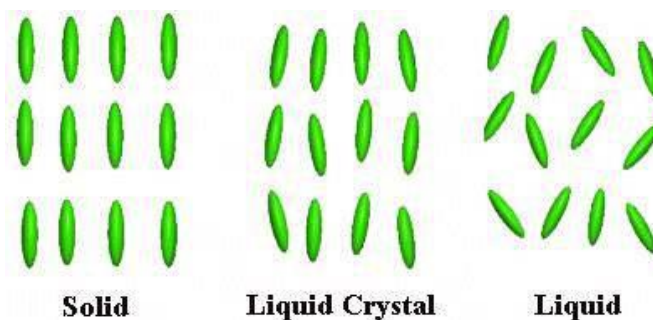


Figure 1: Positional and orientational order in solids, liquids and liquid crystals

In solid crystal state, the molecules are highly ordered. Each molecule occupies a definite position in a more or less rigid arrangement and in immobile rigid arrangement and orient in a preferred direction.

In liquid state, the molecules neither occupy specific positions nor remain oriented in a particular manner. The molecules are overcome with cohesive forces of attraction and are somewhat free to move at random colliding with one another and change their position abruptly.

Intermediate state between the solid state and the liquid state exists another metastable state is called Liquid crystalline phase, wherein the molecules are free to move but are oriented in a particular manner.

For example: Solid cholesteryl benzoate (CB) on heating becomes hazy liquid, which on further heating turns into a clear and transparent liquid. On cooling these changes were observed exactly at the same temperature. Between 145 °C and 179 °C the temperature at which it converts into the liquid crystalline state is called *phase transition temperature*. The temperature at which the transition from liquid crystalline state to isotropic liquid (179 °C) state is called clearing temperature.

The structure of cholesteryl benzoate is given below.

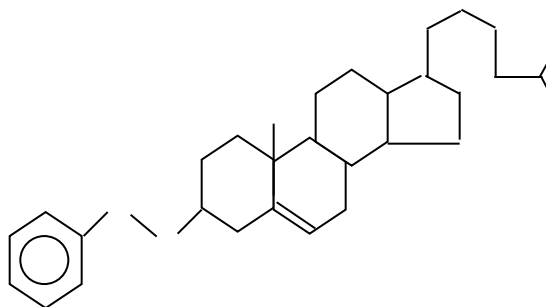
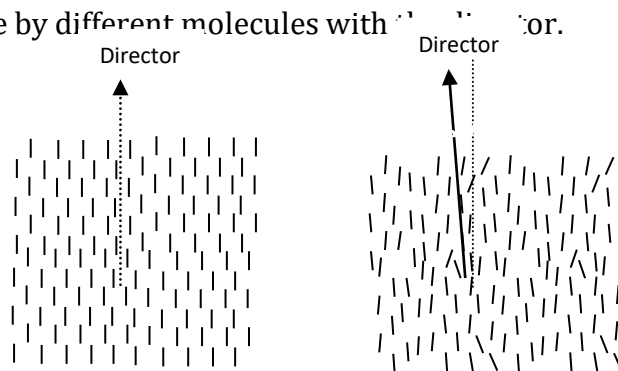


Figure: Structure of cholesteryl benzoate

General applications of liquid crystals: liquid crystal find application in areas of science and engineering, particularly in display systems. Devices using liquid crystalline displays (LCD) have the advantages of low power consumption at lesser cost and hence widely used in electro-optic displays like mobile communication, high-resolution T.V displays, calculator, aircraft cockpit etc. Liquid crystal also finds application in temperature sensors, thermography, pressure sensor etc.

Q- What is Director? Explain its significance.

Director and its significance: In a liquid crystal the molecule passes orientational order i.e. the molecule tends to remain a particular direction. *The direction of preferred orientation of molecules in a liquid crystalline phase is called a director.* Since molecules are in constant motion and they spend more time pointing along the director than any other direction. Hence the extent of orientational order can be described by taking an average angle θ made by different molecules with the director.



Solid state

Liquid crystalline state

Figure: Director in solid state and liquid crystalline state

Significance of director :

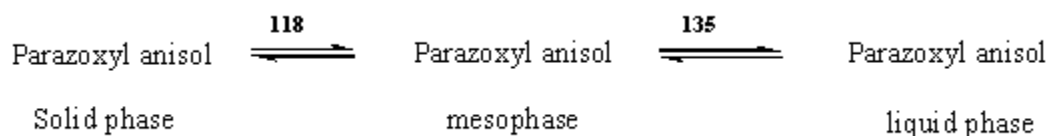
1. An average angle of 0° indicates perfect orientation and can be expected in solid crystals.
2. Liquid crystals exhibit a smaller average angle ($0-5^\circ$) with the director.
3. An average angle greater than 45° indicates no orientational order as found in liquids.

**Q- Discuss the classification of liquid crystals with suitable example
Or Explain the molecular ordering in Thermotropic and Lyotropic liquid crystals.**

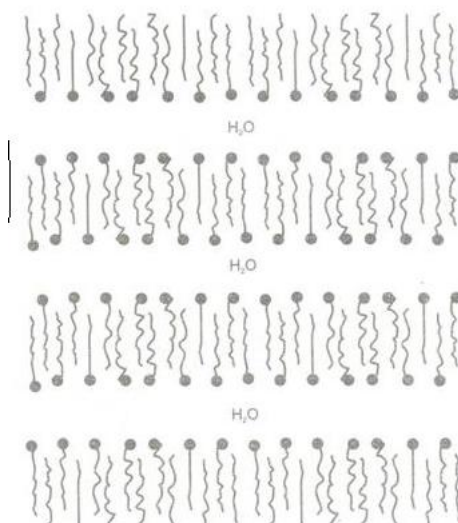
Classification of Liquid crystals: Liquid crystals are classified into two main categories namely

- (i) Thermotropic
- (ii) Lyotropic

(i) Thermotropic liquid crystals: The class of compounds that exhibits crystalline behavior or variation of temperature alone is called thermotropic Lcs. More no. of thermotropic Lcs are formed from organic molecules with a rod like shape.



(ii) Lyotropic liquid crystals: Liquid crystals that are obtained by mixing the two components - organic solids compound (amphiphilic) and polar solvent. Increasing the concentration of the one of the components till liquid phase is obtained such liquid



Figur: Lyotropic mesophase

crystals are called lyotropic liquid crystals. The amphiphilic or amphiphile compound possesses two groups that differ greatly in their solubility properties. One part of the molecule has 'hydrophilic head' which is highly soluble in H_2O . While other part has hydrophobic or lyophobic tail which is highly soluble in hydrocarbon solvents.

In the lyotropic phase, the water is sandwiched between the polar heads of adjacent layers of amphiphilic molecules. Example:

1. Soap water mixture
2. DNA

3. Synthetic polypeptides
4. Biological compounds such as proteins, nucleic acid, also exhibit lyotropic type of liquid crystalline behaviour in water.

Q- What is homologous series? Explain liquid crystalline behavior in homologous PAA homologous series?

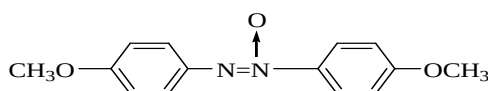
A series of compounds of the same type wherein successive members differ in their molecular formula by CH_2 group is called homologous series. The composition of any homologous series can be expressed by a general formula C_nH_{2n+1} .

The thermal stability of a liquid crystal compound is depending on the structure. Therefore the thermal stability of members of a homologous series changes with change in the number of C-atoms in the chain. On heating a liquid crystalline compound they undergo change from highly ordered arrangement in solid state to disordered isotropic liquid through anisotropic liquid crystalline state. It is interesting to study the phase transition that occurs on heating in homologous series of a liquid crystal.

When the transition temperatures are plotted for a homologous series against the number of C-atoms in the n-alkyl chain of the molecule, certain smooth transition curves are obtained. For a given series, the upper curve represents the transition from liquid crystalline state to the isotropic liquid state (clearing temperature) and the lower curve shows the transition from solid state to liquid crystalline state (transition temperature). The region between two curves gives the range of temperature at which the liquid crystalline state exists.

Generally the first few homologous are purely nematic, the middle will form both nematic and smectic mesophases and the higher homologous does not show significant variation.

PAA homologous series: The liquid crystal of para azoxy anisole (PPA) is stable between 118° and 135° C. PAA has more than 12 homologous which are formed when $-CH_2$ groups are added to its side chain. Plot of the transition temperature against the number of C-atoms in the PAA is shown in fig.



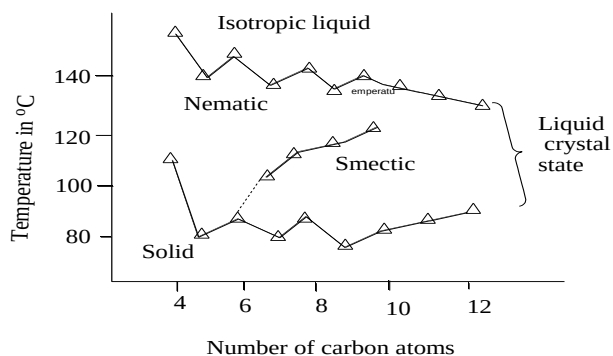


Figure: Phase transition temperature against the number of C-atoms in PAA

From the Figure, the phase transition of PAA is given by

1. When the group (side chain) contains 1-5 C-atoms, the transition is from solid to nematic, nematic to isotropic liquid.
2. Homologous containing 6- 10 C-atoms shows a transition from solid to smectic, smectic to nematic before melting to a liquid.
3. When the number of C-atoms is greater than 10, shows a transition from solid to smectic, smectic to liquid.

Q- Explain the display mechanism in liquid crystals. Or how is liquid utilized in display systems

Display systems

This is used in information display systems, where liquid crystals control the brightness or darkness of parts of a display device. Numbers can be created in numeric displays which have seven segments (also called pixels). Alphabets can be displayed on a fourteen segment arrangement.

A numeric display consists of seven segments for each digit. Each segment is a liquid crystal. Light emitted by each crystal can be independently controlled so that any digit can be displayed. When an electric field is applied to a segment, the liquid crystal is deformed and when polarized light is incident, it gets brightened. By using dichroic dyes, the contrast between bright and dark areas can be enhanced and required colours can be obtained.

One of the most important effects of electrooptic effect (of nematic and chiral liquid crystals) is in liquid display systems (LCD). Electro-optic effect of liquid crystal controls brightness and darkness of emergent light.

LCD requires a very small current. Most of the display in LCD watches and calculators composes of 7 segments. Each seven segments section can display one number. The 14-segments LCD are used of alphabets. Light from the area of each segment are controlled independently and creation of any digits or alphabets is possible. 5x7 matrix segments are used for numbers, letters and figures.

A 7-segment display (Figure) consists of seven segments for each digit (each segment represents a liquid crystal). Light from the area of each of the seven segments or pixels is controlled independently and creation of any of the ten digits becomes possible. When an electric field is applied to a segment, the liquid crystal in that segment undergoes deformation (is activated) and when polarized light is incident, the segment gets brightened. A typical liquid *crystal display cell* (reflection mode) and the electrical circuit with four rows and two columns for a digit to appear is shown in below Figures.

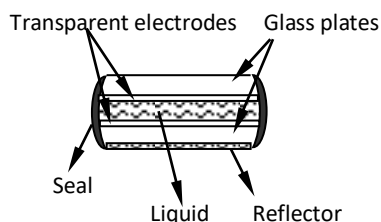
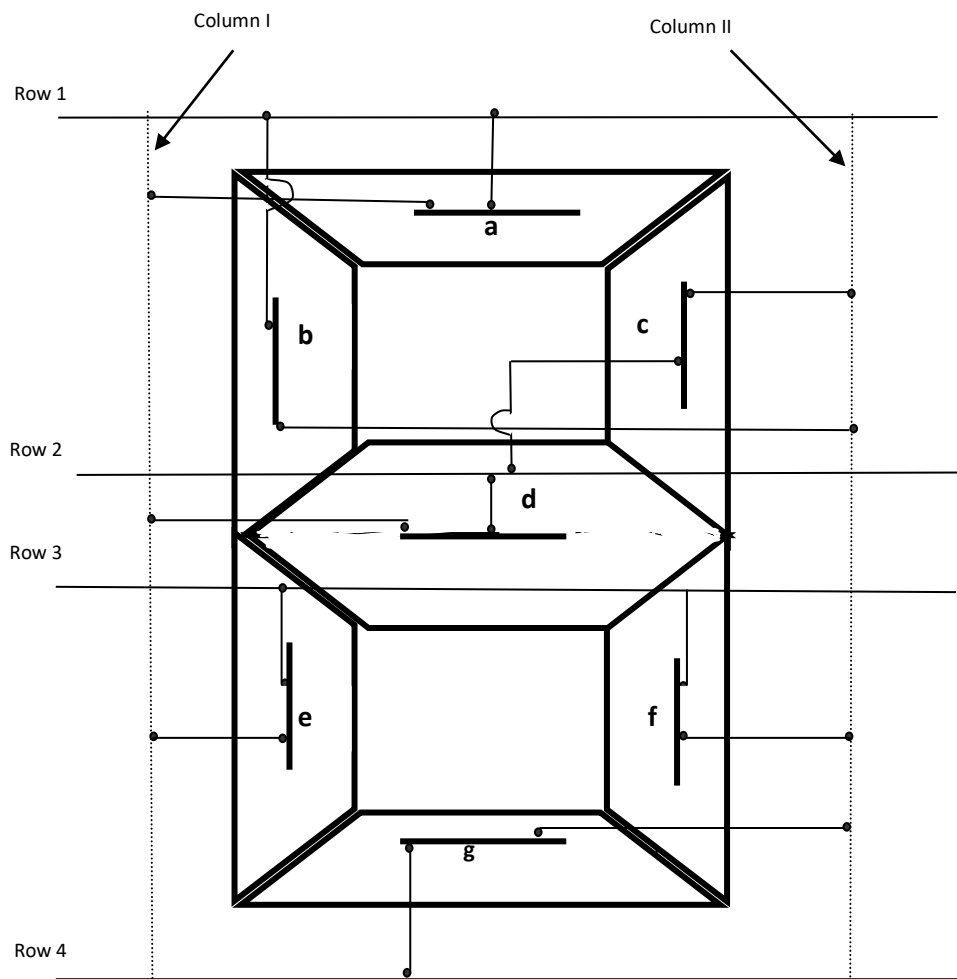


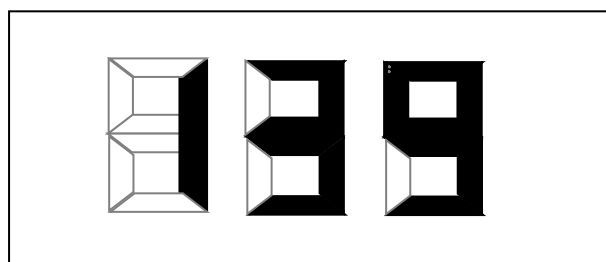
Figure: Representation of a liquid crystal display (reflection mode pixel) where the liquid crystal is embedded between two glass plates

In the Figure, when voltage applied to

- i) Row 1 column I activate the liquid crystal in an area **a** which gets brightened.
- ii) Row 2 column I activates the liquid crystal in an area **d** which gets brightened.
- iii) Row 3 column I activates the liquid crystal in an area **e** which gets brightened.
- iv) Row 1 column II activates the liquid crystal in an area **b** which gets brightened.
- v) Row 2 column II activates the liquid crystal in an area **c** which gets brightened.
- vi) Row 3 column II activates the liquid crystal in an area **f** which gets brightened.
- vii) Row 4 column II activates the liquid crystal in an area **g** which gets brightened.



(a)



(b)

Figure: Seven segment LCD for a digit; each row is connected to two segments and each column is connected to three or four segments; a, to g are the LCD cells. (b) display of number 139

For example, to get a number 139 on the display (Fig. b) an electric field is applied between row 2 and column II (operation v) and between row 3 column II (operation vi) to give '1'

In the second digit operations (i), (v), (vi) and (vii) give '3'

In the third digit operations (i), (ii), (iv), (v), (vi) and (vii) give '9'. Thus the digits 139 appears on the display.

To enhance the difference in the brightness between dark (turned OFF, voltage removed) and bright areas (turned ON, voltage applied), dyes are used. Dyes used in liquid crystal displays are called *dichroic dyes* and give desired colours to the displays with a good contrast.

Display Devices

Organic Light Emitting Diodes (OLED) and their Applications

OLEDs are thin-film organic semiconductor light-emitting devices. also known as organic electroluminescent (organic EL) diode, is a light-emitting diode (LED) in which the emissive electroluminescent layer is a film of organic compound that emits light in response to an electric current. This organic layer is situated between two electrodes; typically, at least one of these electrodes is transparent.

OLED consist of a thin film of organic material sandwiched between two electrodes (anode and cathode). Anode is transparent and made of indium tin oxide. Cathode is reflective and made of metal. When an external potential is applied across electrodes, positive and negative charges (holes from anode and electrons from cathode) are injected. These electrons and holes shift inside the material and recombine to form excitons, emitting photons in the process.

Based on their number of layers, OLEDs can be classified as two-layer and three-layer OLEDs. In two-layer OLEDs, electrons are injected from cathode into the lowest unoccupied molecular orbital. Simultaneously, holes are injected from anode into the highest occupied molecular orbital. The figure given below depicts bilayer PLED

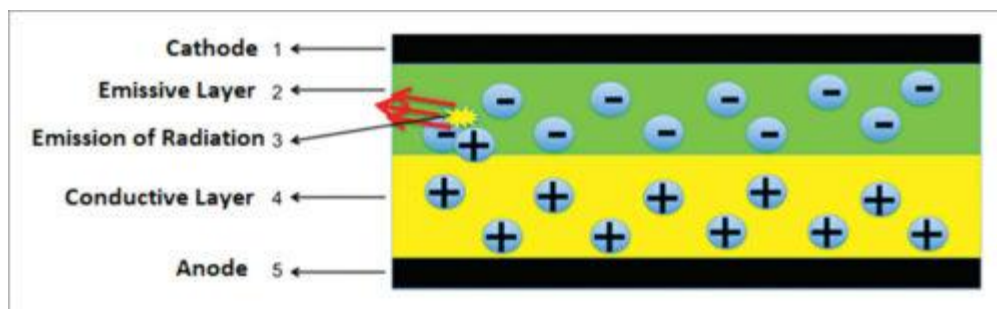


Fig.: Bilayer OLED

An OLED consists of:

- 1. Substrate.** Regarded as the base of an OLED, it is made up of a thin translucent glass or foil material.
- 2. Anode.** It is also called as emitter. Its main function is **to emit electrons when** a voltage is applied across terminals.
- 3. Cathode.** Cathode is the topmost part of **OLED displays. It injects electrons when a potential difference is applied across terminals.**
- 4. Organic layer.** The layer above anode is called organic layer. It contains conductive polymer made of hydrogen or carbon molecules.
- 5. Conductive layer.** This layer is made of organic plastic molecules and helps to move holes from anode.
- 6. Emissive layer.** This layer is made of organic materials, it helps to transport electrons from cathode.

Properties of OLED:

- (i) OLEDs make it easier to fabricate flexible displays.
- (ii) OLEDs fit in as these consume less power and provide high-quality display.
- (iii) The current OLED technology provides remarkable colour fidelity,
- (iv) high efficiency and operation stability
- (v) high reliability and readability.

Applications of OLED: are mainly used

- (i) in digital devices such as high-end television systems,
- (ii) computer monitors,
- (iii) pocket-size systems such as Android phones,
- (iv) media players,

- (v) digital cameras,
- (vi) portable gaming consoles and mini-screens.